

DevOps CI/CD Pipeline With Local Deployment

This is an exercise about building a DevOps CI/CD Pipeline with the following criteria:

Application Setup

- It's a simple Node application with:
 - A frontend
 - A backend
- They are **dockerised** using:
 - a frontend `Dockerfile`
 - a backend `Dockerfile`
 - `docker-compose.yml`

GitHub Actions Workflow

On Merge Request to `dev`:

1. Run linting
Example: `npm run lint`
 2. Run `npm audit`
 3. Run static code security analysis using `snyk`
 4. Execute all tests
 5. Build Docker images
- If all steps pass, allow the merge from `feature` to `dev`

On Merge to `main`:

1. Build production-ready Docker images
2. Copy secrets from GitHub Secrets Manager by creating `.env` files into the image(s)
3. Push the images to Docker Hub:

this local node app will be provided for you

References

- [accessing github secrets](#)
- [eslint](#)
- [snyk](#)